MIL-F-50541A (AR)
8 September 1978
SUPERSEDING
MIL-F-50541 (MU)
15 July 1970

MILITARY SPECIFICATION

FUZE FOR FLECHETTE WARHEAD, WDU-4A/A (LESS EXPULSION CHARGE)

This specification is approved for use by the US Army Armament Research and Development Command, and is available for use by all Departments and Agencies of the Department of Defense.

1. SCOPE

1.1 This specification contains requirements not covered by the drawings and provides quality assurance provisions for the fabrication of parts and assembly of a Fuze for Flechette Warhead, WDU-4A/A (Less Expulsion Charge).

2. APPLICABLE DOCUMENTS

2.1 The following documents of the issue in effect on date of invitation for bids or request for proposals, form a part of this specification to the extent specified herein.

SPECIFICATIONS

MILITARY

MIL-P-38633	-	Primer, Stab, for WDU-4A/A Warhead.
MIL-A-48078	-	Ammunition, Standard Quality Assurance Provisions, General Specification For.
MIL-P-60412	-	Packaging, Packing and Marking For Shipment of Artillery Type and Rocket Fuzes; General Specification For.

Beneficial comments (recommendations, additions, deletions) and any pertinent data which may be of use in improving this document should be addressed to: Commander, US Army Armament Research and Development Command, Attn. DRDAR-QA, Dover, New Jersey 07801 by using the self-addressed Standardization Document Improvement Proposal (DD Form 1426) appearing at the end of this document or by letter.

FSC: 1340

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STANDARDS

MILITARY

MIL-STD-331 - Fuze and Fuze Components, Environmental and Performance Tests For.

MIL-STD-810 - Environmental Test Methods For Aerospace and Ground Equipment.

DRAWINGS (See 6.5)

USAF

65C11046 - Fuze for Flechette Warhead, WDU-4A/A (with expulsion charge)
65C11051 - Fuze for Flechette Warhead, WDU-4A/A (less expulsion charge)
66CI0121 - Container, Packaging
66B10122 - Box, Packing
67D9700 - Warhead 2.75" Rocket, Flechette WDU-4A/A

US Army Research and Development Command (ARRADCOM)

9239406 - Cover
9239407 - Container

(Copies of specifications, standards, drawings and publications required by suppliers in connection with specific procurement functions should be obtained from the procuring activity or as directed by the contracting officer.)

3. REQUIREMENTS

3.1 Material.-Materials shall be in accordance with applicable specifications and drawings.

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3.2 Fuze assemblies and parts.-The fuze assemblies and parts shall comply with all requirements specified on Drawing (dwg) 65C11051, and with all requirements specified in applicable specifications.

3.3 Spring Components

3.3.1 Spring embrittlement.-None of the springs used in the fabrication of the fuze shall break when extended or distorted to the point of minimum permanent distortion.

3.3.2 Spring set.-The coil springs shall return to within the required length of the original free length after compression.

a. Setback weight springs shall return to within .015 inch of the original free length.

b. Firing pin springs shall return to a minimum free length of 1.55 inches immediately after subjection to the load test.

c. Setback sleeve springs shall return to a minimum free length of 2.25 inches immediately after subjection to the load test.

3.4 Safety and Arming Device (S&A)

3.4.1 Non-Arming.-The S&A shall not arm when subjected to an acceleration of 12.75 gravities (g's) minimum for a minimum of 3 seconds.

3.4.2 Arming.-The S&A shall arm in not less than 0.90 seconds and not more than 1.15 seconds when subjected to arming acceleration of 31.0 g's plus or minus 0.25 g's.

3.4.3 Setback weight return capability.-The setback weight in the S&A shall return to the fully safe position after being bottomed with a 20 pound force and released.

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3.4.4 Arming, detent and lock spring functioning.-The rotor assembly in the S&A shall rotate from the fully safe position to the fully armed position, the detent shall lock the rotor in the fully armed position.

3.4.5 Rotor Reset Capability.-The rotor assembly shall be capable of returning to the unarmed position on its own after the S&A device has been partially armed.

3.5 Fuze Assembly Functioning

3.5.1 Non-arming. - The fuze assembly shall not arm when subjected to an acceleration of 15.0 g's plus or minus 0.25 g's for a minimum of 3 seconds.

3.5.2 Arming.-The fuze assembly shall arm in not less than .85 seconds nor more than 1.36 seconds when subjected to an acceleration of 35.0 g's plus or minus one (1) g for a minimum of 1.5 seconds.

3.5.3 Firing.-The fuze assembly after arming in accordance with 3.5 shall fire upon a decrease in acceleration to a maximum of 15.0 g's or below.

3.6 Environment -The fuze assembly shall function in compliance with the requirements of the specification after being subjected to the following environmental conditions.

3.6.1 High Temperature.-The fuze assembly shall function as specified after subjection to High Temperature Procedure I Method 501 of MIL-STD-810.

3.6.2 Low Temperature.-The fuze assembly shall function as specified after subjection to Low Temperature Procedure I at -54°C (-65°F) Method 502 of MIL-STD-810.

3.6.3 Impact Shock.-The fuze assembly shall not arm during and shall function as specified after subjection to a half sine wave shock pulse with a peak of 44 plus or minus 4 g's and time duration of .013 plus or minus .002 seconds.

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3.6.4 Vibration.-The fuze assembly shall function as specified after subjection to Aircraft Vibration, Method 514 as specified in MIL-STD-810 and Appendix A.

3.7 Safety Tests.-The fuze assembly shall remain safe to handle and dispose of after being subjected to the following tests. The disassembled fuze shall display the following conditions:

a. No loose explosive shall be present.

b. The rotor pin shall be in place.

c. The setback sleeve shall be in the safe position.

3.7.1 Jolt.-The fuze assembly shall be subjected to and must withstand the Jolt Test as specified in MIL-STD-331.

3.7.2 Jumble.-The fuze assembly shall be subjected to and must withstand the Jumble Test as specified in MIL-STD-331.

3.7.3 Forty-foot Drop.-The complete fuze assembly shall be subjected to and must withstand the Forty-foot Drop Test as specified in MIL-STD-331.

3.8 First Article Inspection.-This specification contains technical provisions for first article inspection. Requirements for the submission of first article samples by the contractor shall be as specified in the contract.

3.9 Data Requirements.-The contractor shall generate data in accordance with the requirements of the data item description cited in 4.5.18.

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3.10 Workmanship.-All parts and assemblies shall be free from burrs, chips, sharp edges, cracks, unblended radii, surface defects, dirt, grease, rust, porosity, warpage, burn marks, checks, blisters, excess flash, corrosion products and other foreign matter. The cleaning method used shall not be injurious to any part nor shall the parts be contaminated by the cleaning agents. Surface coating shall be continuous except for a few slight scratches not exposing base material The loading and assembly operations shall be performed in a thorough workmanlike manner consistent with the best practices of the industry. All required marking shall be neat and sharply defined.

4. QUALITY ASSURANCE PROVISIONS

4.1 Responsibility for inspection and standard quality assurance provisions.-Unless otherwise specified herein or in the contract, the provisions of MIL-A-48078 shall apply and are hereby made a part of this detail specification.

4.2 Classification of inspection.-The following types of inspection shall be conducted on this item:

a. First Article Inspection

b. Quality Conformance Inspection

4.3 First article inspection.

4.3.1 Submission.-The contractor shall submit a first article sample as designated by the Contracting Officer for evaluation in accordance with provisions of 4.3.2. The first article sample shall consist of the following items in sample quantities as indicated.

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Part description	Drawings	Quantity
Disc, Case	65B11049	25
Lid, Case, Fuze	65C11048	5
Case, Fuze	65C11047	5
Spacer, Fuze	65B11037	5
Container	9239407	5
Cover	9239406	5
Base, Fuze	65C11052	25
Body, Fuze	65C11053	25
Spring, Setback Sleeve	65C11054	25
Pin, Firing, Fuze	65B11055	25
Housing, Arming Mechanism	65C11057	25 after machining
Sleeve, Setback, Fuze	65B11058	25
Fuze for Flechette Warhead, WDU-4A/A (Less Expulsion Charge)	65C11051	204
Spring, Firing Pin	65C11056	25
Spring, Detent	66B10082	25
Spring, Setback Weight	66B10066	25
Gear & Pinion Assembly No 1	8883703	25
Gear & Pinion Assembly No 2	8883700	25
Escapement Gear & Pinion Assembly	8883710	25
Lever Assembly	8883706	30
Rotor Assembly	9239426	20
Inner Plate Assembly	8883693	20
Rotor Housing Assembly	8883719	20
Escapement & Gear Assembly	8883692	20
Safety & Arming Device	9239418	25

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4.3.2 Inspection to be performed.-See MIL-A-48078 and Tables I, II and III specified herein.

4.3.3 Rejection.-See MIL-A-48078.

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TABLE I

FIRST ARTICLE INSPECTION

CLASSIFICATION OF DEFECTS & TESTS

MIL-F-50541A (AR)

PARAGRAPH	TITLE		SHEET 1 OF 6		DRAWING NUMBER
	Fuze for Flechette Warhead, WDU-4A/A (less Expulsion Charge) Assembly and Components				65C11051
					NEXT HIGHER ASSEMBLY
CATEGORY	EXAMINATION OR TEST	NO. OF SAMPLE UNITS	AQL OR 100%	REQUIREMENT PARAGRAPH	PARAGRAPH REFERENCE /INSPECTION METHOD
	Disc, Case (Dwg 65B11049) Examination for Defects	25		3.2	4.3.2.1
	Lid, Case, Fuze (Dwg 65C11048) Examination for Defects	5/ Cavity		3.2	See Table II
	Case, Fuze (Dwg 65C11047) Examination for Defects	5/ Cavity		3.2	See Table III
	Spacer, Fuze (Dwg 65C11037) Examination for Defects	5		3.2	4.3.2.4
	Container (Dwg 9239407) Examination for Defects	5		3.2	4.3.2.5
	Cover (Dwg 9239406) Examination for Defects	5		3.2	4.3.2.6
	Base, Fuze (Dwg 65C11052) Examination for Defects	25		3.2	4.3.2.7
NOTES:					

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TABLE I
FIRST ARTICLE INSPECTION

CLASSIFICATION OF DEFECTS & TESTS

PARAGRAPH	TITLE		SHEET 2 OF 6		DRAWING NUMBER
	Fuze for Flechette Warhead, WDU-4A/A (Less Expulsion Charge) Assembly and Components				65C11051
					NEXT HIGHER ASSEMBLY
CATEGORY	EXAMINATION OR TEST	NO. OF SAMPLE UNITS	AQL OR 100%	REQUIREMENT PARAGRAPH	PARAGRAPH REFERENCE /INSPECTION METHOD
	Body, Fuze (Dwg 65C11053)				
	Examination for Defects	25		3.2	4.4.2.8
	Spring, Setback Sleeve (Dwg 65C11054)				
	Examination for Defects	25		3.2	4.4.2.9
	Spring Embrittlement	10		3.3.1	4.5.2
	Spring Set	15		3.3.2	4.5.3.2
	Spring, Firing Pin (Dwg 65C11056)				
	Examination for Defects	25		3.2	4.4.2.16
	Spring Embrittlement	10		3.3.1	4.5.2
	Spring Set	15		3.3.2	4.5.3.2
	Spring, Detent (Dwg 66B10082)				
	Examination for Defects	25		3.2	4.4.2.17
	Spring Embrittlement	10		3.3.1	4.5.2
	Spring, Setback Weight (Dwg 66B10066)				
	Examination for Defects	25		3.2	4.4.2.18
	Spring Embrittlement	10		3.3.1	4.5.2
	Spring Set	15		3.3.2	4.5.3.1

NOTES:

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TABLE I

FIRST ARTICLE INSPECTION

CLASSIFICATION OF DEFECTS & TESTS

MIL-F-50541A (AR)

PARAGRAPH	TITLE: Fuze for Plechette Warhead, WDU-4A/A (Less Expulsion Charge) Assembly and Components		SHEET 3 OF 6		DRAWING NUMBER: 65C11051
CATEGORY	EXAMINATION OR TEST	NO. OF SAMPLE UNITS	AQL OR 100%	REQUIREMENT PARAGRAPH	NEXT HIGHER ASSEMBLY: PARAGRAPH REFERENCE /INSPECTION METHOD
	Pin, Firing Fuze (Dwg 65B11055)				
	Examination for Defects	25		3.2	4.3.2.10
	Housing, Arming Mechanism (After Machining (Dwg 65C11057)				
	Examination for Defect	25		3.2	4.3.2.11
	Sleeve, Setback, Fuze (Dwg 6581058)				
	Examination for Defect	25		3.2	4.3.2.12
	Fuze for Plechette Warhead, WDU-4A/A Class Expulsion Charge (Dwg 65811051)	204		3.2	4.3.2.19
	Jolt Test	12		3.7.1	4.5.7
	Jumble Test	12		3.7.2	4.5.8
	Forty Foot Drop Test	8		3.7.3	4.5.9
	High Temperature Test	32		3.6.1	4.5.10
	Low Temperature Test	32		3.6.2	4.5.11
	Impact Shock Test	45		3.6.3	4.5.12

NOTES:

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TABLE I

FIRST ARTICLE INSPECTION

CLASSIFICATION OF DEFECTS & TESTS

MIL-F-50541A (AR)

PARAGRAPH	TITLE		SHEET 4 OF 6		DRAWING NUMBER
	Fuze for Flechette Warhead, WDU-4A/A (Less Expulsion Charge) Assembly and Components				65C11051
					NEXT HIGHER ASSEMBLY
CATEGORY	EXAMINATION OR TEST	NO. OF SAMPLE UNITS	AQL OR 100%	REQUIREMENT PARAGRAPH	PARAGRAPH REFERENCE /INSPECTION METHOD
	Non Environmental Fuzes	25		3.2	(see Note A)
	Aircraft Vibration Test	50		3.6.4	4.5.13
	Fuze Non-Arming Test (See Notes A & B)	139		3.5.1	4.5.6.1
	Fuze Arming Time Test (See Notes A & B)	139		3.5.2	4.5.6.2
	Fuze Firing Test (See Notes A & B)	139		3.5.3	4.5.6.3
	Gear and Pinion Assembly No 1 (Dwg 8883703)				
	Examination for Defects	25		3.2	4.4.2.19
	Load Test of No 1 Gear and Pinion Assembly	25		3.2	4.5.4.1
	Gear and Pinion Assembly No 2 (Dwg 8883700)				
	Examination for Defects	25		3.2	4.4.2.20
	Load Test of No 2 Gear and Pinion Assembly	25		3.2	4.5.4.2
	Escapement Gear and Pinion Assembly (Dwg 8883710)				
	Examination for Defects	25		3.2	4.4.2.21
	Load Test of Escapement Gear and Pinion Assembly	25		3.2	4.5.4.3

NOTES: A. 50 fuzes from the aircraft vibration test, 32 fuzes from the high temperature test and 32 fuzes from the low temperature test and 25 non-virornmental fuzes.

B. These tests must be run in the sequence given.

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TABLE I

FIRST ARTICLE INSPECTION

CLASSIFICATION OF DEFECTS & TESTS

MIL-F-50541A (AR)

PARAGRAPH	TITLE	SHEET	DRAWING NUMBER
	Fuze for Flechette Warhead, WDU-4A/A (Less Expulsion Charge) Assembly and Components	5 OF 6	65C11051
			NEXT HIGHER ASSEMBLY

CATEGORY	EXAMINATION OR TEST	NO. OF SAMPLE UNITS	AQL OR 100%	REQUIREMENT PARAGRAPH	PARAGRAPH REFERENCE /INSPECTION METHOD
	Lever Assembly (Dwg 8883706)				
	Examination for Defects	30		3.2	4.4.2.22
	Load Test of Lever Shaft	10		3.2	4.5.4.4
	Load Test of Lever Pallet	10		3.2	4.5.4.5
	Torque Test of Lever Shaft	10		3.2	4.5.4.9
	Rotor Assembly (Dwg 9239426)				
	Examination for Defects	20		3.2	4.4.2.23
	Load Test of Annular Gear	10		3.2	4.5.4.6
	Load Test of Lock Roller Pin	10		3.2	4.5.4.7
	Inner Plate Assembly (Dwg 8883698)				
	Examination for Defects	20		3.2	4.4.2.24
	Load Test of Bridge Rivet	20		3.2	4.5.4.10

NOTES:

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TABLE I

FIRST ARTICLE INSPECTION

CLASSIFICATION OF DEFECTS & TESTS

MIL-F-50541A (AR)

PARAGRAPH	TITLE		SHEET 6 OF 6		DRAWING NUMBER
	Fuze for Flechette Warhead, WDU-4A/A (Less Expulsion Charge) Assembly and Components				65C11051
					NEXT HIGHER ASSEMBLY
CATEGORY	EXAMINATION OR TEST	NO. OF SAMPLE UNITS	AQL OR 100%	REQUIREMENT PARAGRAPH	PARAGRAPH REFERENCE /INSPECTION METHOD
	Rotor Housing Assembly (Dwg 8883719)				
	Examination for Defects	20		3.2	4.4.2.25
	Load Test of Spring Stud	20		3.2	4.5.4.11
	Escapement and Gear Assembly (Dwg 8883692)				
	Examination for Defects	20		3.2	4.4.2.26
	Load Test of Pillars	20		3.2	4.5.4.12
	Safety and Arming Devise (Loading) (Dwg 9239418)				
	Examination for Defects	25		3.2	4.4.2.14
	Rotor Reset Capability	25		3.4.5	4.5.5.6

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TABLE II

FIRST ARTICLE INSPECTION

CLASSIFICATION OF DEFECTS & TESTS

MIL-P-50541A (AR)

PARAGRAPH	TITLE		SHEET 1 OF 1		DRAWING NUMBER 65C11048
	Lid, Case, Fuze				NEXT HIGHER ASSEMBLY 65C11051
CATEGORY	EXAMINATION OR TEST	NO. OF SAMPLE UNITS	AQL OR 100%	REQUIREMENT PARAGRAPH	PARAGRAPH REFERENCE /INSPECTION METHOD
Critical	None Defined				
Major					
101	Outside diameter			3.2	Gage
102	Outside diameter of charge cavity			3.2	Gage
Minor					
201	Outside diameter of flange			3.2	Gage
202	Depth of charge cavity			3.2	Gage
203	Inside diameter of charge cavity			3.2	Gage
204	Thickness of flange			3.2	Gage
205	Perpendicularity of outside diameter to charge cavity flange			3.2	Gage
206	Evidence of poor workmanship			3.10	Visual

NOTES:

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TABLE III

FIRST ARTICLE INSPECTION

CLASSIFICATION OF DEFECTS & TESTS

MIL-F-50541A (AR)

PARAGRAPH	TITLE		SHEET 1 OF 1		DRAWING NUMBER 65C11047
	Case, Fuze				NEXT HIGHER ASSEMBLY 65C11051
CATEGORY	EXAMINATION OR TEST	NO. OF SAMPLE UNITS	AQL OR 100%	REQUIREMENT PARAGRAPH	PARAGRAPH REFERENCE /INSPECTION METHOD
Critical	None Defined				
Major					
101	Inside diameter at open end of case			3.2	Gage
102	Inside diameter at base of case			3.2	Gage
103	Depth of case			3.2	Gage
104	Concentricity of inside diameter at open end of case with outside diameter flange			3.2	Gage
105	Outside diameter at lid interface			3.2	Gage/Visual
106	Vent holes missing or improperly located in case			3.2	Gage/Visual
Minor					
201	Fins missing or deformed			3.2	Gage
202	Evidence of poor workmanship			3.10	Visual

NOTES:

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4.4 Quality conformance inspection

4.4.1 Inspection lot formation - Inspection lots shall comply with the lot formation provisions of MIL-A-48078.

4.4.2 Examination - See MIL-A-48078.

a. Sampling plans. - Unless otherwise specified in the classification of Defects and Test tables, sampling plans and procedures for major and minor defects shall be in accordance with MIL-STD-105, Inspection Level II.

b. Process specification for plastic parts. - Prior to the start of normal production, the contractor shall obtain approval from the procurng activity for the process specification of plastic parts submitted with the first article sample. This shall specify the time temperature and pressure ranges which shall be maintained during production of the molded parts. At unscheduled intervals during the operation of the approved process, the Government inspector shall observe the process to ascertain that the approved limits are being maintained. No deviation from these limits without approval from the procuring activity shall be allowed. In the event of an approved deviation, the contractor shall submit a new first article sample.

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CLASSIFICATION OF DEFECTS & TESTS

PARAGRAPH	TITLE		SHEET 1 OF 1		DRAWING NUMBER 65C11049
4.4.2.1	Disc, Case				NEXT HIGHER ASSEMBLY 65C11046
CATEGORY	EXAMINATION OR TEST	NO. OF SAMPLE UNITS	AQL OR 100%	REQUIREMENT PARAGRAPH	PARAGRAPH REFERENCE /INSPECTION METHOD
Critical	None Defined				
Major	None Defined				
Minor					
201	Diameter, max		0.65%	3.2	Gage
202	Evidence of poor workmanship		0.65%	3.10	Visual

NOTES:

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QUALITY CONFORMANCE INSPECTION

CLASSIFICATION OF DEFECTS & TESTS

MIL-F-50541A (AR)

PARAGRAPH	TITLE		SHEET 1 OF 1		DRAWING NUMBER 65C11048
4.4.2.2	Lid, Case, Fuze				NEXT HIGHER ASSEMBLY 65C11046
CATEGORY	EXAMINATION OR TEST	NO. OF SAMPLE UNITS	AQL OR 100%	REQUIREMENT PARAGRAPH	PARAGRAPH REFERENCE /INSPECTION METHOD
Critical	None Defined				
Major					
101	Outside Diameter		0.40%	3.2	Gage
102	Outside Diameter of Charge Cavity		0.40%	3.2	Gage
Minor					
201	Inside Diameter of Charge Cavity		0.65%	3.2	Gage
202	Depth of Charge Cavity		0.65%	3.2	Gage
203	Thickness of Flange		0.65%	3.2	Gage
204	Evidence of Poor Workmanship		0.65%	3.10	Visual

NOTES:

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CLASSIFICATION OF DEFECTS & TESTS

MIL-F-50541A (AR)

PARAGRAPH	TITLE		SHEET 1 OF 1		DRAWING NUMBER
4.4.2.3	Case, Fuze				65C11047
					NEXT HIGHER ASSEMBLY 65C11046
CATEGORY	EXAMINATION OR TEST	NO. OF SAMPLE UNITS	AQL OR 100%	REQUIREMENT PARAGRAPH	PARAGRAPH REFERENCE /INSPECTION METHOD
Critical	None Defined				
Major					
101	Inside Diameter at open end of case		0.40%	3.2	Gage
102	Inside Diameter at base of case		0.40%	3.2	Gage
103	Depth of case		0.40%	3.2	Gage
Minor					
201	Outside diameter of flange		0.65%	3.2	Gage
202	Evidence of poor workmanship		0.65%	3.10	Visual

NOTES:

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QUALITY CONFORMANCE INSPECTION

CLASSIFICATION OF DEFECTS & TESTS

MIL-F-50541A (AR)

PARAGRAPH	TITLE		SHEET 1 OF 1		DRAWING NUMBER
4.4.2.4	Spacer, Fuze				65B11037
					NEXT HIGHER ASSEMBLY 65C11046
CATEGORY	EXAMINATION OR TEST	NO. OF SAMPLE UNITS	AQL OR 100%	REQUIREMENT PARAGRAPH	PARAGRAPH REFERENCE /INSPECTION METHOD
Critical	None Defined				
Major	None Defined				
Minor					
201	Diameter		0.65%	3.2	Gage
202	Thickness		0.65%	3.2	Gage
203	Evidence of poor workmanship		0.65%	3.10	Visual

NOTES:

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QUALITY CONFORMANCE INSPECTION

CLASSIFICATION OF DEFECTS & TESTS

MIL-F-50541A (AR)

PARAGRAPH	TITLE		SHEET 1 OF 1		DRAWING NUMBER 9239407
4.4.2.5	Container				NEXT HIGHER ASSEMBLY 65C11046
CATEGORY	EXAMINATION OR TEST	NO. OF SAMPLE UNITS	AQL OR 100%	REQUIREMENT PARAGRAPH	PARAGRAPH REFERENCE /INSPECTION METHOD
Critical	None Defined				
Major					
101	Outside diameter		0.40%	3.2	Gage
102	Diameter of charge cup, max		0.40%	3.2	Gage
103	Height		0.40%	3.2	Gage
Minor					
201	Evidence of poor workmanship		0.65%	3.10	Visual

NOTES:

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CLASSIFICATION OF DEFECTS & TESTS

MIL-F-50541A (AR)

PARAGRAPH	TITLE		SHEET 1 OF 1		DRAWING NUMBER 9239406
4.4.2.6	Cover				NEXT HIGHER ASSEMBLY 65C11046
CATEGORY	EXAMINATION OR TEST	NO. OF SAMPLE UNITS	AQL OR 100%	REQUIREMENT PARAGRAPH	PARAGRAPH REFERENCE /INSPECTION METHOD
Critical	None Defined				
Major 101	Diameter		0.40%	3.2	Gage
Minor 201	Evidence of poor workmanship		0.65%	3.10	Visual

NOTES:

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CLASSIFICATION OF DEFECTS & TESTS

MIL-F-50541A (AR)

PARAGRAPH	TITLE		SHEET 1 OF 1		DRAWING NUMBER
4.4.2.7	Base, Fuze				65C11052
					NEXT HIGHER ASSEMBLY 65C11051
CATEGORY	EXAMINATION OR TEST	NO. OF SAMPLE UNITS	AQL OR 100%	REQUIREMENT PARAGRAPH	PARAGRAPH REFERENCE /INSPECTION METHOD
Critical	None Defined				
Major					
101	Outside diameter		0.40%	3.2	Gage
102	Inside diameter, Max		0.40%	3.2	Gage
103	Concentricity of outside diameter with inside diameter		0.40%	3.2	Gage
Minor					
201	Evidence of poor workmanship		0.65%	3.10	Visual

NOTES:

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CLASSIFICATION OF DEFECTS & TESTS

MIL-P-50541A (AR)

PARAGRAPH	TITLE		SHEET 1 OF 1		DRAWING NUMBER
4.4.2.8	Body, Fuze				65C11053
					NEXT HIGHER ASSEMBLY
					65C11051
CATEGORY	EXAMINATION OR TEST	NO. OF SAMPLE UNITS	AQL OR 100%	REQUIREMENT PARAGRAPH	PARAGRAPH REFERENCE /INSPECTION METHOD
Critical	None Defined				
Major					
101	Small outside diameter, min		0.40%	3.2	Gage
102	Inside diameter of firing pin cavity		0.40%	3.2	Gage
103	Diameter of holes for locking balls, min		0.40%	3.2	Gage
104	Inside diameter of S&A cavity, max		0.40%	3.2	Gage
105	Evidence of burrs on holes for locking balls and vent holes		0.40%	3.2	Gage
106	Length from S&A seating surface to center of locking ball holes		0.40%	3.2	Gage
107	Perpendiculairty of locking ball holes to small outside diameter		0.40%	3.2	Gage
108	Finish improper		0.40%	3.2	Gage
Minor					
201	Evidence of poor workmanship		0.65%	3.10	Visual

NOTES:

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CLASSIFICATION OF DEFECTS & TESTS

PARAGRAPH	TITLE		SHEET 1 OF 1		DRAWING NUMBER 65C11054
4.4.2.9	Spring, Setback Sleeve				NEXT HIGHER ASSEMBLY 65C11051
CATEGORY	EXAMINATION OR TEST	NO. OF SAMPLE UNITS	AQL OR 100%	REQUIREMENT PARAGRAPH	PARAGRAPH REFERENCE /INSPECTION METHOD
Critical	None Defined				
Major					
101	Inside diameter		0.40%	3.2	Gage
102	Load at compressed length		0.40%	3.2	Gage
103	Minimum length after load test		0.40%	3.2	Gage
104	Spring embrittlement	10		3.3.1	4.5.2
105	Spring set	10		3.3.2	4.5.3.2
Minor					
201	Evidence of poor workmanship		0.65%	3.10	Visual

NOTES:

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CLASSIFICATION OF DEFECTS & TESTS

MIL-F-50541A (AR)

PARAGRAPH	TITLE		SHEET 1 OF 1		DRAWING NUMBER 65B11055
4.4.2.10	Pin, Firing				NEXT HIGHER ASSEMBLY 65C11051
CATEGORY	EXAMINATION OR TEST	NO. OF SAMPLE UNITS	AQL OR 100%	REQUIREMENT PARAGRAPH	PARAGRAPH REFERENCE /INSPECTION METHOD
Critical	None Defined				
Major					
101	Large outside diameter		0.40%	3.2	Gage
102	Diameter of firing pin point		0.40%	3.2	Gage
103	Taper angle of firing pin point		0.40%	3.2	Gage
Minor					
201	Small outside diameter		0.65%	3.2	Gage
202	Overall length		0.65%	3.2	Gage
203	Length to base of firing pin point		0.65%	3.2	Gage
204	Length to tapered shoulder between large and small outside diameter		0.65%	3.2	Gage
205	Angle of tapered shoulder		0.65%	3.2	Gage
206	Evidence of poor workmanship		0.65%	3.10	Visual

NOTES:

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CLASSIFICATION OF DEFECTS & TESTS

MIL-F-50541A (AR)

PARAGRAPH	TITLE		SHEET 1 OF 1		DRAWING NUMBER 65C11057
4.4.2.11	Housing, Arming Mechanism				NEXT HIGHER ASSEMBLY 65C11051
CATEGORY	EXAMINATION OR TEST	NO. OF SAMPLE UNITS	AQL OR 100%	REQUIREMENT PARAGRAPH	PARAGRAPH REFERENCE /INSPECTION METHOD
Critical	None Defined				
Major					
101	Diameter of flange		0.40%	3.2	Gage
102	Length from rear face of flange to base of tabs		0.40%	3.2	Gage
103	Length from forward face offlange to base surface of tabs		0.40%	3.2	Gage
104	Diameter of body mating surface, min		0.40%	3.2	Gage
Minor					
201	Presence of flash hole		0.65%	3.2	Gage
202	Evidence of poor workmanship		0.65%	3.10	Visual

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CLASSIFICATION OF DEFECTS & TESTS

MIL-P-50541A (AR)

PARAGRAPH	TITLE		SHEET 1 OF 1		DRAWING NUMBER
4.4.2.12	Sleeve, Setback, Fuze				65B11058
					NEXT HIGHER ASSEMBLY
					65C11051
CATEGORY	EXAMINATION OR TEST	NO. OF SAMPLE UNITS	AQL OR 100%	REQUIREMENT PARAGRAPH	PARAGRAPH REFERENCE /INSPECTION METHOD
Critical	None Defined				
Major					
101	Large outside diameter		0.40%	3.2	Gage
102	Small inside diameter		0.40%	3.2	Gage
103	True Position of large outside diameter with small inside diameter		0.40%	3.2	Gage
Minor					
201	Evidence of poor workmanship		0.65%	3.10	Visual

NOTES:

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CLASSIFICATION OF DEFECTS & TESTS

MIL-F-50541A (AR)

PARAGRAPH	TITLE		SHEET 1 OF 1		DRAWING NUMBER
4.4.2.13	Safety and Arming Device (metal parts)				9239423
					NEXT HIGHER ASSEMBLY
					65C11051
CATEGORY	EXAMINATION OR TEST	NO. OF SAMPLE UNITS	AQL OR 100%	REQUIREMENT PARAGRAPH	PARAGRAPH REFERENCE /INSPECTION METHOD
Critical					
1	Distance between annular gear and and No. 2 gear, max		100%	3.2	Gage
Major					
101	Length end of rotor pin, max		0.40%	3.2	Gage
102	Gear train completely engaged		0.40%	3.2	Visual/Manual
Minor					
201	Evidence of poor workmanship		0.65%	3.10	Visual

NOTES:

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QUALITY CONFORMANCE INSPECTION
CLASSIFICATION OF DEFECTS & TESTS

PARAGRAPH	TITLE		SHEET 1 OF 1		DRAWING NUMBER 9239418
4.4.2.14	Safety and Arming Device (Loading)				NEXT HIGHER ASSEMBLY 65C11051
CATEGORY	EXAMINATION OR TEST	NO. OF SAMPLE UNITS	AQL OR 100%	REQUIREMENT PARAGRAPH	PARAGRAPH REFERENCE /INSPECTION METHOD
Critical					
1	Non-arming test		100%	3.4.1	4.5.5.1
2	Setback weight return capability		100%	3.4.3	4.5.5.3
3	Rotor reset capability		100%	3.4.5	4.5.5.2
4	Load test of rotor pin		100%	3.2	4.5.4.8
Major					
101	Height to top of rotor, max		0.40%	3.2	Gage
102	Primer improperly staked into place		0.40%	3.2	Visual
103	Foreign matter present		0.40%	3.2	Visual
104	Arming, detent functioning test		100%	3.4.4	4.5.5.4
105	Arming test		100%	3.4.2	4.5.5.5
106	Torque test of rotor stop stud (Eccentric Pin)		0.40%	3.2	4.5.4.13
107	Evidence that S&A is armed or partially armed	(b)	100%	3.2	Visual
Minor					
201	Evidence of poor workmanship		0.65%	3.10	Visual

NOTES: (a)Primers and Detonators are to be tested in accordance with para 4.5.14
(b)To be performed after all other examinations and tests.

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QUALITY CONFORMANCE INSPECTION

CLASSIFICATION OF DEFECTS & TESTS

PARAGRAPH	TITLE	SHEET	DRAWING NUMBER
4.4.2.15	Fuze For Flechette Warhead, WDU-4A/A (Less Expulsion Charge)	1 OF 1	65C11051
			NEXT HIGHER ASSEMBLY

CATEGORY	EXAMINATION OR TEST	NO. OF SAMPLE UNITS	AQL OR 100%	REQUIREMENT PARAGRAPH	PARAGRAPH REFERENCE /INSPECTION METHOD
Critical					
1	Jolt Test (See Note C)	12		3.7.1	4.5.7
2	Jumble Test (See Note C)	12		3.7.2	4.5.8
3	Impact Shock Test	45		3.6.3	4.5.12
4	Fuze Non-Arming Test (See Notes A&B)	75		3.5.1	4.5.6.1
Major					
101	Length from base to forward end of fuze body, max		0.40%	3.2	Gage
102	Gap between fuze body and S&A housing, max		0.40%	3.2	Gage
103	Firing pin and firing pin spring in improper position		0.40%	3.2	Visual
104	Foreign matter present		0.40%	3.2	Visual
105	Rotor pin in improper position		0.40%	3.2	Visual
106	Aircraft Vibration Test, MIL-STD-810, Test Method 514, Curve - See Appendix A	50	0.40%	3.2 3.6.4	4.5.13
107	Fuze Arming Test (See notes A&B)	75		3.5.2	4.5.6.2
108	Fuze Firing Test (See notes A&B)	75		3.5.3	4.5.6.3
Minor					
201	Marking missing or incomplete		0.65%	3.2	Visual
202	Evidence of poor workmanship		0.65%	3.10	Visual

NOTES: A. 50 fuzes from the aircraft vibration test plus 25 non-environmental fuzes shall be used for these tests. B. These test must be run in the sequence given: Fuze non- arming, fuze arming, fuze firing. C. The Jumble Test shall immediately follow the Jolt Test using the same fuzes. Fuzes from the Jolt Test shall not be disassambled until after the Jumble Test.

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CLASSIFICATION OF DEFECTS & TESTS

MIL-P-50541A (AR)

PARAGRAPH	TITLE		SHEET 1 OF 1		DRAWING NUMBER 65C11056
4.4.2.16	Spring, Firing Pin				NEXT HIGHER ASSEMBLY 65C11051
CATEGORY	EXAMINATION OR TEST	NO. OF SAMPLE UNITS	AQL OR 100%	REQUIREMENT PARAGRAPH	PARAGRAPH REFERENCE /INSPECTION METHOD
Critical	None Defined				
Major					
101	Load at compressed length		0.40%	3.2	Gage
102	Minimum length after Load Test		0.40%	3.2	Gage
Minor					
201	Evidence of poor workmanship		0.65%	3.10	Visual

NOTES:

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PARAGRAPH	TITLE		SHEET 1 OF 1		DRAWING NUMBER 66B10082
4.4.2.17	Spring, Detent				NEXT HIGHER ASSEMBLY 65C11051
CATEGORY	EXAMINATION OR TEST	NO. OF SAMPLE UNITS	AQL OR 100%	REQUIREMENT PARAGRAPH	PARAGRAPH REFERENCE /INSPECTION METHOD
Critical	None Defined				
Major 101	Spring embrittlement	10		3.3.1	4.5.2
Minor 201	Evidence of poor workmanship		0.65%	3.10	Visual

NOTES:

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CLASSIFICATION OF DEFECTS & TESTS

PARAGRAPH	TITLE		SHEET 1 OF 1		DRAWING NUMBER 65B11067
4.4.2.18	Spring, Setback Weight				NEXT HIGHER ASSEMBLY 65C11051
CATEGORY	EXAMINATION OR TEST	NO. OF SAMPLE UNITS	AQL OR 100%	REQUIREMENT PARAGRAPH	PARAGRAPH REFERENCE /INSPECTION METHOD
Critical	None Defined				
Major					
101	Spring Embrittlement	10		3.3.1	4.5.2
102	Spring Set	10		3.3.2	4.5.3.1
Minor					
201	Evidence of poor workmanship		0.65%	3.10	Visual

NOTES:

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CLASSIFICATION OF DEFECTS & TESTS

MIL-F-50541A (AR)

PARAGRAPH	TITLE		SHEET 1 OF 1		DRAWING NUMBER
4.4.2.19	Gear and Pinion Assembly No. 1				8883703
					NEXT HIGHER ASSEMBLY 8883692
CATEGORY	EXAMINATION OR TEST	NO. OF SAMPLE UNITS	AQL OR 100%	REQUIREMENT PARAGRAPH	PARAGRAPH REFERENCE /INSPECTION METHOD
Critical					
1	Load Test of No. 1 Gear and Pinion Assembly		100%	3.2	4.5.4.1 Gage
2	Maximum gap between gear and pinion		100%	3.2	
Major	None Defined				
Minor 201	Evidence of poor workmanship		0.65%	3.10	Visual

NOTES:

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CLASSIFICATION OF DEFECTS & TESTS

MIL-F-50541A (AR)

PARAGRAPH	TITLE		SHEET 1 OF 1		DRAWING NUMBER 8883700
4.4.2.20	Gear and Pinion Assembly No. 2				NEXT HIGHER ASSEMBLY 8883692
CATEGORY	EXAMINATION OR TEST	NO. OF SAMPLE UNITS	AQL OR 100%	REQUIREMENT PARAGRAPH	PARAGRAPH REFERENCE /INSPECTION METHOD
Critical 1	Load Test of No. 2 Gear and Pinion Assembly		100%	3.2	4.5.4.2 Gage
	Maximum gap betwwen gear and pinion		100%	3.2	
Major	None Defined				
Minor 201	Evidence of poor workmanship		0.65%	3.10	Visual

NOTES:

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CLASSIFICATION OF DEFECTS & TESTS

MIL-F-50541A (AR)

PARAGRAPH	TITLE		SHEET 1 OF 1		DRAWING NUMBER 8883710
4.4.2.21	Escapement Gear and Pinion Assembly				NEXT HIGHER ASSEMBLY 8883692
CATEGORY	EXAMINATION OR TEST	NO. OF SAMPLE UNITS	AQL OR 100%	REQUIREMENT PARAGRAPH	PARAGRAPH REFERENCE /INSPECTION METHOD
Critical 1	Load Test of No. 2 Gear and Pinion Assembly		100%	3.2	4.5.4.3 Gage
	Maximum gap between gear and pinion		100%	3.2	
Major	None Defined				
Minor 201	Evidence of poor workmanship		0.65%	3.10	Visual

NOTES:

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CLASSIFICATION OF DEFECTS & TESTS

MIL-P-50541A (AR)

PARAGRAPH	TITLE		SHEET 1 OF 1		DRAWING NUMBER 8883706
4.4.2.22	Lever Assembly				NEXT HIGHER ASSEMBLY 8883692
CATEGORY	EXAMINATION OR TEST	NO. OF SAMPLE UNITS	AQL OR 100%	REQUIREMENT PARAGRAPH	PARAGRAPH REFERENCE /INSPECTION METHOD
Critical					
1	Load test of lever shaft		100%	3.2	4.5.4.4
2	Load test of lever pallet		100%	3.2	4.5.4.5
3	Maximum gap between lever and shaft		100%	3.2	Gage
Special					
A	Torque test of lever shaft (See Note)		0.040%	3.2	4.5.4.9
Major	None Defined				
Minor					
201	Evidence of poor workmanship		0.65%	3.10	Visual
	NOTE: The sampling plan for this test in accordance with MIL-STD-105 using inspection Level II and an AQL of 0.040 per cent. Continuous sampling may be used in accordance with MIL-STD-1235 using inspection Level II, CSP-I and an AQL of 0.015 per cent.				

NOTES:

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CLASSIFICATION OF DEFECTS & TESTS

PARAGRAPH	TITLE		SHEET		DRAWING NUMBER
4.4.2.23	Rotor Assembly		1 OF 1		9239426
					NEXT HIGHER ASSEMBLY 65C11046
CATEGORY	EXAMINATION OR TEST	NO. OF SAMPLE UNITS	AQL OR 100%	REQUIREMENT PARAGRAPH	PARAGRAPH REFERENCE /INSPECTION METHOD
Critical 1	Load test of annular gear		100%	3.2	4.5.7.6
Special A	Load test of lock roller pin (See Note)		0.040%	3.2	4.5.7.7
Major	None Defined				
Minor 201	Evidence of poor workmanship		0.65%	3.10	Visual
	NOTE: The sampling plan for this test in accordance with MIL-STD-105 using inspection Level II and an AQL of 0.040 per cent. Continuous sampling may be used in accordance with MIL-STD-1235 using inspection Level II, CSP-I and an AQL of 0.015 per cent.				

NOTES:

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CLASSIFICATION OF DEFECTS & TESTS

MIL-P-50541A (AR)

PARAGRAPH	TITLE		SHEET 1 OF 1		DRAWING NUMBER 8883693
4.4.2.24	Inner Plate Assembly				NEXT HIGHER ASSEMBLY 65C11046
CATEGORY	EXAMINATION OR TEST	NO. OF SAMPLE UNITS	AQL OR 100%	REQUIREMENT PARAGRAPH	PARAGRAPH REFERENCE /INSPECTION METHOD
Critical	None Defined				
Special A	Load Test of bridge rivet (See Note)		0.040%	3.2	4.5.4.10
Major	None Defined				
Minor 201	Evidence of poor workmanship		0.65%	3.10	Visual
	NOTE: The sampling plan for this test in accordance with MIL-STD-105 using inspection Level II and an AQL of 0.040 per cent. Continuous sampling may be used in accordance with MIL-STD-1235 using inspection Level II, CSP-1 and an AQL of 0.015 per cent.				

NOTES:

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CLASSIFICATION OF DEFECTS & TESTS

MIL-F-50541A (AR)

PARAGRAPH	TITLE		SHEET 1 OF 1		DRAWING NUMBER
4.4.2.25	Rotor Housing Assembly				8883719
					NEXT HIGHER ASSEMBLY 65C11046
CATEGORY	EXAMINATION OR TEST	NO. OF SAMPLE UNITS	AQL OR 100%	REQUIREMENT PARAGRAPH	PARAGRAPH REFERENCE /INSPECTION METHOD
Critical	None Defined				
Major					
101	Load test of spring stud		0.40%	3.2	4.5.4.11
102	Rotor housing arm width		100%	3.2	Gage
Minor					
201	Evidence of poor workmanship		0.65%	3.10	Visual

NOTES:

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CLASSIFICATION OF DEFECTS & TESTS

MIL-F-50541A (AR)

PARAGRAPH	TITLE		SHEET		DRAWING NUMBER
4.4.2.26	Escapement and Gear Assembly		1 OF 1		8883692
					NEXT HIGHER ASSEMBLY
					65C11056
CATEGORY	EXAMINATION OR TEST	NO. OF SAMPLE UNITS	AQL OR 100%	REQUIREMENT PARAGRAPH	PARAGRAPH REFERENCE /INSPECTION METHOD
Critical	None Defined				
Major 101	Load test of pillars		0.40%	3.2	4.5.4.12
Minor 201	Evidence of poor workmanship		0.65%	3.10	Visual

NOTES:

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4.4.3 Testing - Testing is described in the First Article and Quality Conformance Inspection Tables.

4.4.4 Inspection Equipment - The inspection equipment required to perform the examinations and tests prescribed herein is described in the "Paragraph Reference/Inspection Method" column in the tables starting with paragraph 4.4.2.1. The contractor shall submit for approval inspection equipment designs in accordance with the terms of the contract. See Section 6 of MIL-A-48078 and section 6.3 herein.

4.4.4.1 Contractor Inspection System - As part of contractor's approved quality procedure, there shall be a provision for checking the adequacy of inspections made in connection with listed critical or major defects. These procedures shall include the following provisions:

4.4.4.1.1 Handling Procedure - A Design Agency approved procedure shall be instituted that shall preclude unauthorized handling of the S&A's and fuzes from the S&A position check. The procedure shall be fail safe to eliminate armed fuzes from being introduced into the lot. Fail safe is defined as follows: Any equipment failure shall operate to cause rejection of the part or otherwise prevent its subsequent use.

4.4.4.1.2 Identification of Rejected Material - All material listed below, rejected as containing defects, shall be isolated to preclude its inclusion with acceptable material:

a. All S&A's failing paragraphs 4.5.5.1, 4.5.5.3, 4.5.5.4 4.5.5.5, 4.5.5.6.

b. All fuzes failing paragraphs 4.5.7 and 4.5.8, 4.5.9, 4.5.13, 4.5.6.2

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4.4.4.2 Use of Simulated Defect Standards - The following listed verified inspections shall be validated and inspection equipment verified hourly using Design Agency approved item facsimilies or standards containing the defects and supplied by the contractor. The contractor shall resupply the item facsimilies when it is determined by the Government that the facsimilies are no longer representative enough of the product to make it unidentifiable to the contractor inspection personnel, or it is not functioning in the necessary manner and cannot be corrected satisfactorily. These standards should be inert and readily identifiable to the Government to ensure recovery should they be inadvertently introduced into production. Simulated defect standards used to verify the following verified inspections should be constructed and identified in such a manner that they are not readily recognizable to the contractor inspection personnel as defect standards. Standards should be used so that their number, time and sequence of introduction into the inspection cycle is known only to those personnel responsible for checking the accuracy of the operation. Adequacy of the standards shall be verified at least once each shift.

Verified Inspection

Paragraph 4.5.5.3 - Setback weight return capability.

NOTE: When there is automatic detection and automatic rejection of defective items to a controlled area or a controlled rejection of defective items, it is permissible for the defect standards to be identifiable and their introduction into production noticeable to contractor inspection personnel.

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4.4.4.3 Undetected Defect Standards - Provisions shall be established for the corrective action to be taken where the equipment or contractor inspection personnel fail to identify a standard as defective. These provisions shall include a reinspection of all acceptable material subsequent to the last acceptable verification check as a minimum.

4.5 Methods of Inspection

PRECAUTION: This specification covers sampling and testing of chemical, toxic or explosive materials which are potentially hazardous to personnel. Accordingly, it is emphasized that all applicable safety rules, regulations and procedures must be followed in handling and processing these materials.

4.5.1 Materials, components and processes - Compliance with all requirement of Section 3 of this specification shall be ascertained by current and continuing examination of inspection and test data to determine that all components (parts, subassemblies and materials) have been inspected and tested and found to comply with their respective drawing and specification requirements, and that all specified manufacturing processes have been followed.

4.5.1.1 Visual and Mechancial Inspection - The visual, manual and mechanical inspection and tests shall verify compliance with the requirements of Section 3 and 5 of this specification in accordance with 4.4.2 herein and Tables I, II, III.

4.5.2 Spring embrittlement. - The spring shall be stretched along its longitudinal axis, in the case of a coil spring, or bent in the case of a flat spring, until the spring no longer returns to its original position. When the spring no longer returns to its original length or profile, it shall be considered distorted permanently. Any spring that breaks prior to this point shall be classed defective. If one or more springs fail to comply with the specified requirements the spring lot shall be rejected.

4.5.3 Load testing of springs

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4.5.3.1 Setback weight spring. - The springs shall be measured at their original free length, assembled in a restraining fixture in accordance with 4.3.4 and compressed to a normal height of .750 inches. The fixture shall be placed in an oven and subjected to a temperature of 175 + 5 degrees F for 24 hours. The fixture shall be removed from the oven and be cooled to ambient temperature for 24 hours, minimum. The springs shall then be released and measured to determine the change in free length in accordance with the requirement. Any spring that does not meet the free length requirement after the test shall be classed defective. If one or more springs fail to comply with the specified requirements the the spring lot shall be rejected.

4.5.3.2 Setback sleeve and firing pin springs. - Each spring shall be tested with a spring tester for the required load at the height required by the drawing, .975 inches for the setback sleeve spring and .600 inches for the firing pin spring. Immediately after release from the spring tester, each spring shall be gaged for recovery to the minimum required free length. Any spring that does not meet the free length requirement after the test shall be classed defective. If one or more springs fail to comply to the specified requirements the spring lot shall be rejected.

4.5.4 Mechanical Tests

4.5.4.1 Load test of No. 1 gear and pinion asembly. - Apply an axial load perpendicular to the No. 1 gear and in such a direction as to test the assembly of the No. 1 gear and pinion. The resistive force shall act against the No. 1 gear. Upon removal of the required force, the maximum permissible gap shall be inspected for compliance with the drawing requirement. Any gear and pinion assembly that breaks or does not meet the maximum gap requirement shall be classed defective.

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4.5.4.2 Load test of No. 2 gear and pinion assembly. - Apply an axial load perpendicular to the No. 2 gear and in such a direction as to test the assembly of the No. 2 gear and pinion. The resistive force shall act against the No. 2 gear. Upon removal of the required force, the maximum permissible gap shall be inspected for compliance with the drawing requirement. An gear and pinion assembly that breaks or does not meet the maximum gap requirement shall be classed defective.

4.5.4.3 Load test of escapement gear and pinion assembly.- Apply an axial load perpendicular to the escapement gear and in such a direction as to test the assembly of the escapement gear and pinion. The resistive force shall act against the escapement gear. Upon removal of the required force, the maximum permissible gap shall be inspected for compliance with the drawing requirement. Any gear and pinion assembly that breaks or does not meet the maximum Gap requirements shall be classed defective.

4.5.4.4 Load test of lever shaft. - Apply an axial load perpendicular to the lever and in such a direction as to test the assembly of the lever and lever shaft. The resistive force shall act against the lever. Upon removal of the required force, the maximum permissible gap shall be inspected for compliance with the drawing requirement.

4.5.4.5 Load test of lever pallet. - Apply an axial load perpendicular to the lever and in such a direction as to test the assembly of the lever and the lever pallet. The resistive force shall act against the lever. Upon removal of the required force, for compliance with the drawing requirement. Any lever pallet that does not comply with the drawing requirement shall be classed defective.

4.5.4.6 Load test of annular gear. - Apply a load perpendicular to the rotor and in such a direction as to test the assembly of the annular gear studs to the annular gear and the rotor. The resistive force shall act against the rotor.

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Upon removal of the required force, the distance from the face of the rotor to the outside face of the annular gear shall be inspected for compliance with the drawing requirement. Any annular gear that does not comply with the drawing requirement shall be classed defective.

4.5.4.7 Load test of lock roller pin. - Apply an axial load perpendicular to the rotor and in such a direction as to test the strength of the assembly of the lock roller pin to the rotor. The resistive force shall act against the rotor. Any lock roller pin that will not support the load shall be classed defective.

4.5.4.8 Load test of rotor pin. - Apply a load perpendicular to the rotor and in such a direction as to test the assembly of the rotor pin to the rotor. The resistive force shall act against the rotor. Any rotor pin that will not support the load shall be classed defective.

4.5.4.9 Torque Test of lever Shaft. - A Torque shall be applied between the lever and the lever shaft. Any lever shaft that will not support the torque shall be classed defective.

4.5.4.10 Load test of bridge rivet. - Apply the required axial load perpendicular to the inner plate and into the bridge rivet from the side opposite the bridge. The resistive force shall act against the inner plate. Any bridge rivet that will not support the load shall be classed defective.

4.5.4.11 Load test of spring stud. - Apply an axial load perpendicualr to the rotor housing and in such a direction as to test the assembly of the spring stud and the rotor housing. The resistive force shall act against the rotor housing. Any spring stud that will not support the load shall be classed defective.

4.5.4.12 Load test of pillars. - Apply the required axial load perpendicular to the inner plate and in such a direction as to test the assembly of the pillars to the

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inner plate and outer plate. The resistive force shall act against the inner plate. Any pillar that will not support the load shall be classed defective.

4.5.4.13 The torque test of rotor stop stud (eccentric pin).- The required torque shall be applied between the rotor stop stud and the gear assembly. The resistive force shall act on the gear assembly. Any rotor stop stud that will not withstand the required torque shall be classed defective.

4.5.5 S&A Tests

4.5.5.1 Non-arming. - The loaded S&A with the g-weight in the downward position shall be assembled to the test equipment in accordance with 4.3.4. The equipment shall accelerate the loaded S&A in a clockwise direction to 12.75 g's minimum for a minimum of 3 seconds. Observation shall be made for rotor in the safe position. Any S&A that is found armed or partially armed shall be classed defective.

4.5.5.2 Rotor reset capability. - The loaded S&A shall be assembled to the test fixture in accordance with 4.3.4. The rotor shall be rotated to a position a minimum of 25 degrees but not more than 35 degrees from the fully safe position and released. Observation shall be made for rotor return to the fully safe position. Any rotor that fails to return to the safe position shall be classed defective.

4.5.5.3 Setback return capability. - The loaded S&A shall be assembled to the test fixture in accordance with 4.3.4. A force of 20 lbs shall be applied equally to the two top surfaces of the setback weight toward the base of the rotor housing, and then released. Observation shall be made for setback weight return to the fully safe position. If ary setback weight fails to return to the fully safe position, the S&A shall be classified defective.

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4.5.5.4 Arming, detent functioning. - The loaded S&A in the safe position shall be assembled to the test equipment in accordance with 4.3.4. The equipment shall accelerate the loaded S&A to 27.50 g's maximum for a minimum of 3 plus 1, minus 0 seconds in a direction which simulates rocket launch. At the completion of acceleration, the equipment shall be halted and visual observation shall be made for the locking position of the detent. Finger tip pressure shall be applied to the rotor to rotate in the safe direction to ascertain positive arming. At the completion of the test the loaded S&A shall be returned to the safe position. Any S&A that fails to detent shall be classed defective.

4.5.5.5 Arming. - The loaded S&A with the g-weight in the downward position shall be assembled to the test equipment in accordance with 4.3.4. The equipment shall accelerte the loaded S&A in a clockwise direction to 31.00 g's plus or minus 0.25 g's and the arming time shall be recorded to the nearest hundredth of a second using a calibrated timer. Any S&A that fails arming time shall be classed defective.

4.5.6 Fuze assembly functioning

4.5.6.1 Non-arm. - The fuze assembly shall be assembled to the test equipment in accordance with 4.3.4. The equipment shall accelerate the fuze assembly to 15.00 g's plus or minus 0.25 g's for a minimum of 3 seconds. Each fuze assembly shall be examined for safe position of rotor pin, set-back sleeve, and firing pin. Any fuze that fails non-arming shall be classed defective.

4.5.6.2 Arming. - The fuze assembly shall be assembled to the test equipment in accordance with 4.3.4. The equipment shall accelerate the fuze to 35.0 g's plus or minus one (1) g for a minimum of 1.5 seconds. The arming time shall be recorded to the nearest hundredth of a second using a swing fixture and acoustic pick-up feeding the arming cycle noise signal to a recorder.

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The fuze assembly will normally fire upon decrease in acceleration to 15.0 g's or below after arming. Provision shall be made to prevent a fuze from firing when the unit has been observed as an arming time defect. The S&A shall then be returned to the safe position before disassembly of the fuze from the fixture. Each fuze shall be examined to determine if the S&A has detented properly, failure to detent shall be classed as an arming time defect. (See para 4.5.15 and Table IV). Fuzes which have not fired when the centrifuge is stopped shall be examined for safe position before handling. A fuze which has evidently armed but has not fired shall be classified as a failure to fire and handled with an approved safety method to prevent injury.

4.5.6.3 Firing. - The fuze assemby after being armed in accordance with 4.4.5.2 will release the firing pin and detonate the primer when the acceleration has reduced to 15.0 g's or below. Each fuze shall be visually examined to verify that the primer has fired. Any fuze that has not fired shall be classed defective. (See Table IV)

4.5.7 Jolt - The fuze assembly shall be attached to the jolt machine by means of a Government approved fixture which simulates enclosure in the Warhead, 2.75 Inch Rocket, Flechette WDU - 4A/A (See dwg 67D9700). The fuze assembly in this confinement shall be subjected to and pass the Jolt Test as specified in MIL-STD-331. The same fuze assembly shall than be subjected to the Jumble Test in paragraph 4.5.8. before being disassembled. If one or more fuze assemblies fail to meet the test requirements, the lot shall be rejected.

4.5.8 Jumble - The fuze assembly shall be subjected to and pass the Jumble Test as specified in MIL-STD-331 using a Government approved fixture which simulates enclosure in the Flechette Warhead. If one or more fuze assemblies fails to meet the test requirements, the lot shall be rejected.

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4.5.9 Forty foot drop - The fuze assembly shall be assembled into the loaded fuze configuration (see dwg. 65C11046) and assembled into warhead, 2.75 Inch Rocket, Flechete WDU-4A/A loaded to live weight (see dwg. 67D9700). The warhead shall be packaged (see dwg 66C10121), packed (see dwg 66B10122) and subjected to and pass the Forty-foot test as specified in MIL-STD-331. The packed box shall be dropped base down only. If one or more fuzes are found to be in an unsafe condition the lot shall be rejected.

4.5.10 High temperature - The fuze assembly shall be subjected to and pass high temperature environment in accordance with High Temperture Procedure I, method 501 of MIL-STD-810 with the exception that the time of exposure to temperature shall be a minimum of four (4) hours. The fuze shall be armed and fired within 5 minutes of removal from extreme temperature.

4.5.11 Low temperature - The fuze assembly shall be subjected to and pass low temperature environment in accordance with Low Temperature Procedure I at-54°C (-65°F). Method 502 of MIL-STD-810 with the exception that the time of exposure to temperature shall be a minimum of four (4) hours. The fuze shall be armed and fired withn 5 minutes of removal from extreme temperature.

4.5.12 Impact shock - The fuze assembly shall be attached to a fixture, base down, and subjected to a half sine wave shock pulse with a peak of 44 plus or minus 4 g's and time duration of .013 plus or minus .002 seconds. The shock pulse test method shall be monitored with a modified fuze having a rigidly attached accelerometer, with suitable frequency response, to the base of the S&A rotor housing. The impact shock shall be repeated a total of three (3) times for each fuze. The fuze assembly shall remain safe and shall not fire. Any fuze found to be not safe or has fired shall be classed defective.

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4.5.13 Vibration - The fuze assembly shall be attached to the vibration machine by means of a Government approved fixture which simulates enclosure in the Warhead, 2.75 Inch Rocket, Flechette WDU-4A/A (see dwg 67D9700). The fuze assembly in this confinement shall be subjected to Aircraft Vibration in accordance with MIL-STD-810, method 514, Curve-See appendix A except that resonance shall not be done. Any fuze that does not meet vibration requirements shall be classed defective.

4.5.14 Check Test for Deterioration of Primers - If the total time between original acceptance of any primer lot and the assembly of that lot into the fuzes exceeds two years, at any time since previous test, the primer lot shall be subjected to, and must satisfactorily pass the functioning and output tests specified in MIL-P-38633 immediately before the primer lot is assembled into the fuzes.

4.5.15 Production Acceptance Criteria - The sample fuzes from the aircraft vibration and non-environmental tests shall be combined into one sample and subjected to the following tests specified in Table IV.

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TABLE IV

SAMPLE SIZE - 75

TEST	ACCEPT	REJECT	TEST METHOD PARAGRAPH
Non-arm	0	1	4.5.6.1
Arming Time under min.	0	1	4.5.6.2
	Point Assessment		
Arming Time Over max.		2 points	
Fail to arm		4 points	

The fuze lot shall be rejected if the fuze assemblies do not meet the above criteria or have a total point accumulation of 7 points or more.

NOTE: An Arming time over max. failure is two points while a fail to arm failure is four points. The total points based on the failures determines the disposition of the lot (see 4.5.16).

If two (2) or more fuzes fail to fire upon deceleration during production testing, the lot shall be rejected.

4.5.16 Fuze Arming and Non-Arming Definitions -

a. Times under .85 seconds are classed as arming time under minimum.

b. Times 1.37 seconds through 1.42 seconds are classed as arming time over maximum.

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c. Times over 1.42 seconds or fuzes which do not arm or fuzes which time but do not detent are classed as failures to arm.

d. Identification of defects as non-arm failures, arming time under minimum, arming time over maximum, and failures to arm will depend on both centrifuge testing testng and radiographic examination.

e. If the test equipment fails to produce an arming readout (i.e., recording paper skewes, audio pick-up or amplifier inoperative) during the arming time test as verified by the Government, the Safety and Arming Device shall be removed from the fuze body; and if found to be armed or partially armed, the Safety and Arming Device shall be reset into the unarmed position, reassembled into the same fuze body and retimed once as a fuze. If the fuze has fired and arming time is lost, that fuze will be discounted for the arming time test only. Hence, the fuze arming test will contain one less sample. The only handling action performed on the Safety and Arming Device shall be the reset of the device just prior to the non-arming and arming time tests. The timing record produced by the fuze on the approved fuze timing equipment shall be considered the timing record of the fuze under the applicable defect classification.

f. Fuze arming times, if recorded to the thousandth of a second shall be "rounded off" to the nearest hundredth of a second. Example:

(1) 0.845 shall be 0.84
0.846 shall be 0.85
(2) 1.365 shall be 1.36
1.366 shall be 1.37
(3) 1.415 shall be 1.42

4.5.17 Correlation Factor - A correlation factor, which correlates the fuze arming time method (swing fixture) to the in-line S&A arming time method, shall be run for this fuze. Correlation tests shall be performed by the contractor with DCAS verification. Correlation factors shall be validated at six month intervals (maximum) and re-established whenever:

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a. A new contract is issued
b. A change of fuze type
c. Fuze timing equipment is modified

For the correlation procedure a digital counter or an approval equivalent device shall be used, set to read milliseconds for S&A timing and set for oscillograph traces, to the nearest millisecond, for the fuze. The correlation shall be accomplished as follows:

a. Select five (5) unloaded S&A's demonstrating acceptable arming time. Each S&A shall have repeatable arming times within 10 milliseconds in five successive runs on the approved S&A timing equipment. S&A arming times readings shall be obtained from the first movement of the set-back weight to the detent locking in the rotor.

b. The same S&A's shall be placed in fuze housing. The assemblies shall then be timed five (5) times on the approved fuze timing equipment. Readings of the fuze arming time traces, from set back weight bottoming to rotor detent, shall be recorded.

c. The shift in average fuze time, if any, when compared with the average S&A time (taken as a standard) shall be used as the correlation correction and applied to the acceptance test results.

Data obtained in accordance with the above procedure shall be submitted to Commander, US Army Research and Development Command, Dover, NJ 07801, ATTN: DRDAR-QAR-M and DRDAR-QAR-I.

4.5.18 The contractor shall submit a quality inspection test, demonstration and evaluation report in accordance with Data Item Description DI-R-1721 and DI-R1724 giving the detailed test results.

5. Preparation for Delivery

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5.1 Packaging and marking - Level C - The fuze, less expulsion charge, shall be packaged in accordance with MIL-P-60412. Marking shall be on labels.

5.2 Packing and marking - Level C - The packaged fuzes shall be packed and marked in accordance with MIL-P-60412. Marking shall be on labels. The ICC nomenclature shall be "Percussion Fuzes".

6. Notes

6.1 Intended use - The components covered by this specification are intended for use on the 2.75 Inch Rocket.

6.2 Ordering data - See MIL-A-48078

6.3 Submission of inspection equipment for design approvals - See MIL-A-48078. Submit designs as required to Commander, US Army Armament Research and Development Command, ATTN: DRDAR-QAR-I, Dover, NJ 07801

6.4 Goverment furnished parts

6.4.1 Packing and packaging - The procuring agency shall provide the packing and packaging for the Forty-foot Drop Test.

6.4.2 WDU-4A/A Warheads - The procuring agency shall provide the inert WDU-4A/A warhead for the Forty-foot Drop Test.

6.5 Drawings - Drawings listed in Section 2 of this specification under the heading US Army Armament Research and Development Command (ARRADCOM) may also include drawings prepared by, and identified as Edgewood Arsenal, Frankford Arsenal, Rock Island Arsenal or Picatinny Arsenal drawings. Technical data originally prepared by these activities is now under the cognizance of ARRADCOM.

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6.6 Defect Reporting - From each lot produced, the contractor shall submit two copies of the Metal Parts Lot Acceptance Test Data (See Appendix B) to ARRADCOM, ATTN: DRDAR-QAR-M and DRDAR-LCN, within three (3) days after completion of acceptance testing of the lot. If fuzes and/or tests, in addition to acceptance requirements, are utilized to evaluate the lot, the results of these tests will also be submitted three (3) days after the additional tests are completed. The defective fuzes shall be saved pending review of the results. Recommended test data sheet is enclosed.

6.7 Changes from Previous Issue - Asterisks are not used in this revision to identify changes with respect to the previous issue, due to the extensiveness of the changes.

6.8 Data Requirements - Deliverable data required by this specification in the following paragraphs:

Paragraph	Data Requirements	Applicable DID
4.5.18	Quality Inspection Test, Demonstration and Evaluation Report	DI-R-1724
4.5.18	Quality Inspection Defect Report	DI-R-1721

(Copies of data item description required by the contractor in connection with specific procurement functions should be obtained from the procuring activity or as direction by the contracting officer).

Custodian:
Army-Ar

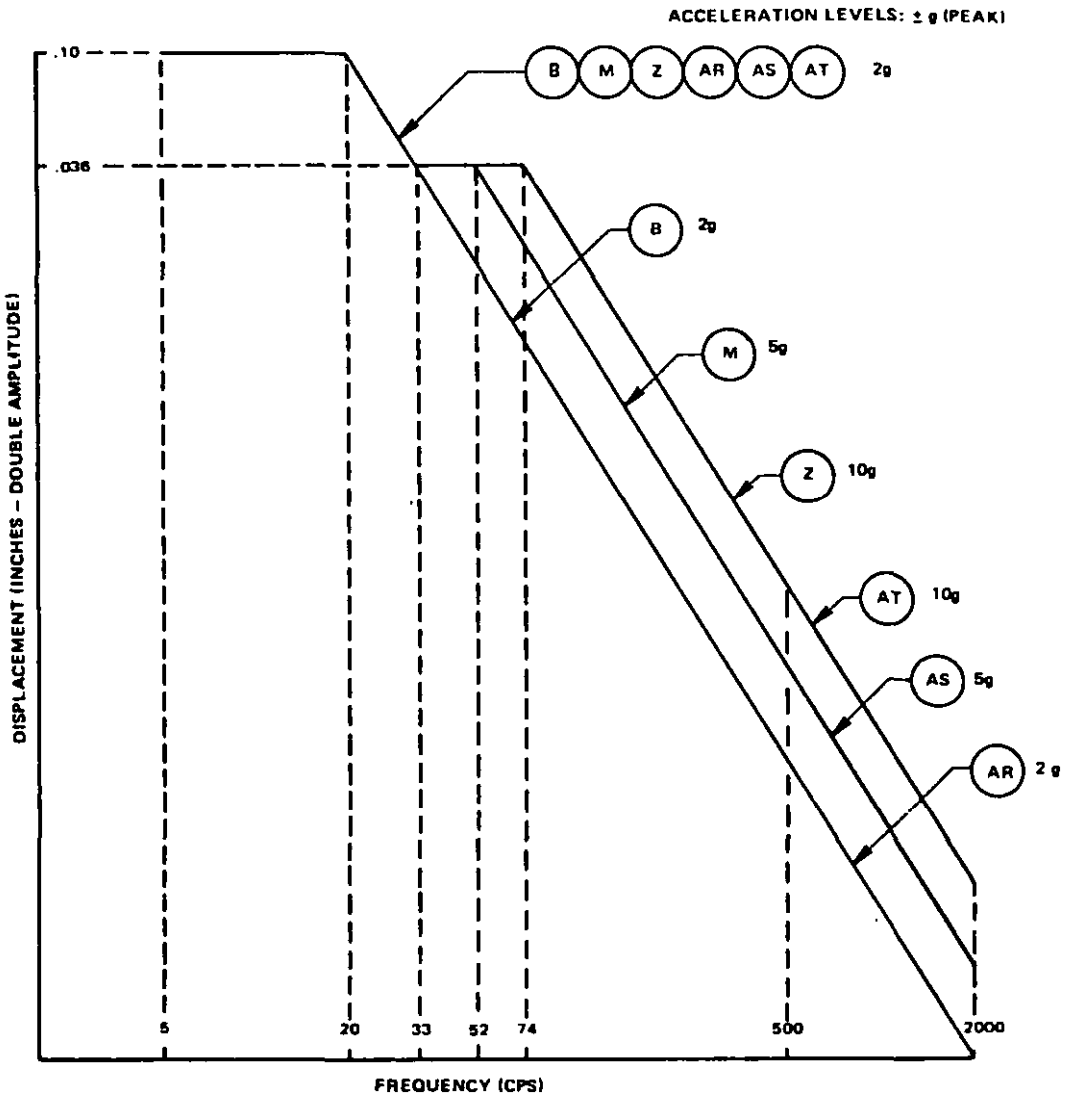
Preparing Activity
Army-AR

Project Number: 1340-A-547

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APPENDIX A

Curve Z shall be used for the Helicopter Vibration Test. See paragraph 3.6.4.



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APPENDIX A (continued)

A. Test procedure and time schedule chart for equipment installed in airplanes and helicopters – equipment category (a)

Equipment mounting configuration	Procedure number	Procedure part number	Applicable tests (see 4 for test procedures)			Test time schedule (per axis)				Curve (note 1)
			Resonance search (4.5.1.1)	Resonance dwell (4.5.1.2)	Sinusoidal cycling (4.5.1.3)	Dwell time at each resonance (4.5.1.2)	Sinusoidal cycling time (4.5.1.3)	Sweep time 5-500-5 cps	Sweep time 5-2000-5 cps	
Without vibration isolators	1	1	X	X	X	30 min	3-hrs-less dwell time	15 min	20 min	M,Z,AS,AT
With vibration isolators (note 2)	1	1	X	X	X	30 min	3-hrs-less dwell time	15 min	20 min	M,Z,AS,AT
		2	X	X	X	10 min	30 min	15 min	20 min	B,AR
Normally with vibration isolators but tested without isolators	1	2	X	X	X	10 min	30 min	15 min	20 min	B,AR

Note 1: For sinusoidal vibration resonance tests and cycling tests of items mounted in airplanes and helicopters and weighing more than 80 pounds, the vibratory accelerations shall be reduced by +/- 1 g for each 20-pound increment of weight over 80 pounds. However, the vibratory acceleration shall in no case be less than 50 percent of the specified curve level.

Note 2: Test items of equipment normally provided with vibration isolators first shall be tested with the isolators in place (Part 1). The isolators then shall be removed, and the test item rigidly mounted and subjected to the test level indicated. (Part 2).

B. Curve selection chart for category (a) equipment

Selection criteria	Curve (for freq. to 500 cps)	Curve (for freq. to 2000 cps such as jet aircraft)
Equipment designed for both helicopter and airplane applications when the location in the airplane is in the forward half of fuselage or in wing areas of airplanes with engines at rear of fuselage.	M	AS
Equipment designed for both helicopter and airplane applications when the location in the airplane is in the rear half of the fuselage or in wing areas of airplanes with wing or front mounted engines or other equipment or engine locations not specifically mentioned for other curves.	Z	AT
Equipment installed on vibration isolated panels or racks when the panel or rack is *not available for test or when the equipment is tested with isolators removed* as specified by the applicable procedure.	B	AR

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APPENDIX B

M_____METAL PARTS LOT ACCEPTANCE TEST DATA

Contractor:_____ Fuze Lot No:_____

Date of Test Lot Quantity (Gross):_____

Start:_____ Sample Size (Total):_____

End:_____ Sample Size Returned to Lot:_____

Date Shipped:_____ Lot Quantity (Net):_____

Operational Test

Non-Arm Test

Sample Size	Serial No of Defects	Failure Analysis

Arming Time Test

Sample Size	Number Accepted	Serial No of Short Times	Serial No of long Times	Timed But Failed to Arm	Not Time Failed to Arm	Armed But Not Timed

Attach list of fuze serial numbers and arming times.

Total Defect Points:_____

Correlation Factor applied to the arming time reading of the Operational Test samples was_____seconds.

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Firing Test

Sample Size	Serial No of Defects	Failure Analysis

Vibration Test

Sample Size	Serial No of Defects	Failure Analysis

Failure Analysis

If lot failure is due to other than test conditioning

Government Inspector - Date

Contractor Representative - Date

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DOCUMENT IDENTIFIER *(Number)* AND TITLE
MIL-F-50541A (AR) - Fuze for Flechete Warhead WDU-4A/A(Less Expulsion Charge)

NAME OF ORGANIZATION AND ADDRESS OF SUBMITTER

☐ VENDOR ☐ USER ☐ MANUFACTURER

1. ☐ HAS ANY PART OF THE DOCUMENT CREATED PROBLEMS OR REQUIRED INTERPRETATION IN PROCUREMENT USE? ☐ IS ANY PART OF IT TOO RIGID, RESTRICTIVE, LOOSE OR AMBIGUOUS? PLEASE EXPLAIN BELOW.

A. GIVE PARAGRAPH NUMBER AND WORDING

B. RECOMMENDED WORDING CHANGE

C. REASON FOR RECOMMENDED CHANGE(S)

2. REMARKS

SUBMITTED BY *(Printed or typed name and address — Optional)*

TELEPHONE NO.

DATE

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